Human-in-the-Loop Labor Architectures: Reinstating and Reconfiguring Work through Skills, Certification, and Human-Centered AI

V. M. Benignus Grero

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Abstract

AI-enabled automation is reshaping production, services, and knowledge work, bringing real concerns about displacement, deskilling, and the concentration of decision power in opaque systems. At the same time, research in human–AI interaction, human-in-the-loop (HITL) methods, and explainable AI (XAI) shows that AI can be designed to augment human capabilities and support work practices that are more equitable and trustworthy. Existing work, however, tends to focus on interaction patterns and system-level governance, while leaving the design of jobs, skills, and certification pathways under-specified. This paper introduces the concept of *human-in-the-loop labor architectures*: socio-technical configurations in which HITL and XAI capabilities are deliberately used to reinstate and reconfigure jobs, define new skill profiles, and anchor certification and policy interventions. We propose a four-layer framework—capabilities, roles, skills and certification, and governance—that connects AI affordances to occupational structures and institutional arrangements. Drawing on research on AI and the future of work, human-centered AI, and organisational fairness, we show how this architecture can reinstate and upgrade roles in automation, human resources, collaboration, and well-being, and how governments and institutions can use it to elevate human capabilities and quality of life in an evolving global economy.

Keywords: human-in-the-loop (HITL); explainable AI (XAI); future of work; human–AI interaction; labor architecture; capability layer; role layer; skills and certification layer; governance layer; job reinstatement; job reconfiguration; human-centered AI; algorithmic governance; workforce development; occupational skills; certification pathways; AI oversight; human–AI co-performance; organisational fairness; digital well-being.

1 Introduction

1.1 The AI Transformation of Work

Artificial intelligence has moved from the periphery to the core of organisational infrastructure. Across sectors, AI systems increasingly shape decisions about hiring, promotion, performance management, task allocation, and process control. In manufacturing, cyber-physical systems and

machine-learning controllers optimise production schedules, predict maintenance needs, and adjust workflows in real time. In human resources, predictive models estimate candidate competence, screen applications, and guide internal mobility decisions. In digital collaboration, AI tools monitor participation patterns, analyse team dynamics, and suggest interventions. In workplace well-being, AI-based monitoring systems assess workload, stress levels, and employee engagement through passive data collection and behavioural analysis.

These systems promise major efficiency gains, cost reductions, and new capabilities. Yet they also raise concerns that cut across technical, organisational, and human dimensions. Workers and job applicants report unease about opaque and potentially biased AI decisions in recruitment and performance evaluation. Researchers document increased mental workload and stress associated with digital tools and automated systems, particularly when users lack meaningful control over how these systems operate. Organisations struggle to define accountability when AI systems mediate high-stakes decisions affecting people's livelihoods and working conditions. Policymakers face mounting pressure to ensure fairness, transparency, and worker protection in an increasingly AI-mediated economy.

1.2 The Polarised Public Discourse

Public discourse on AI and work often polarises around two scenarios. On one side, prognosticators warn of widespread technological unemployment, with AI systems displacing workers across entire occupational categories. On the other side, optimists argue that AI will prove to be a net job creator, generating new roles through innovation, productivity growth, and the emergence of previously unimaginable industries and services.

Both narratives share a critical limitation: they largely treat job structures and role definitions as passive outcomes of technological change. The implicit assumption is that technology arrives, and labour markets adjust—either painfully or productively. What remains under-theorised is the possibility that the design of AI systems themselves, and the institutional arrangements surrounding their deployment, can actively shape whether work is displaced, downgraded, or elevated.

1.3 The Promise of Human-Centered AI

A growing body of research in human-AI interaction and human-centered AI offers an alternative perspective. This work demonstrates that the design of AI systems—how they expose information, support feedback, allocate control, and render their operations intelligible—profoundly shapes how humans participate in and experience AI-mediated work.

Human-in-the-loop (HITL) methods embed humans into data collection, model training, and decision-making loops, enabling them to evaluate, correct, or guide AI behaviour. Rather than treating humans as mere recipients of AI outputs, HITL approaches position them as active participants whose judgment is structurally necessary for system performance. Explainable AI (XAI) techniques provide stakeholders with intelligible information about model reasoning, uncertainty, and limitations. By making AI systems more transparent, XAI enables humans to understand, contest, or appropriately rely on model outputs, creating the conditions for meaningful human oversight. Human-centered AI frameworks argue for configurations that combine high levels of computer automation with high levels of human control. Such configurations aim to achieve reliability, safety,

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and trustworthiness by ensuring that humans can understand machine states, assess consequences, and intervene when appropriate.

Tsiakas and Murray-Rust extend these ideas specifically to future-of-work practices, identifying a design space for HITL and XAI across four domains: automation, human resources, collaboration, and human factors. They emphasise co-performance and distributed responsibility, showing how humans and AI systems can work together in mutually enhancing ways.

1.4 The Missing Element: Labor Architectures

Yet even in this valuable work, jobs and occupational structures remain largely implicit. Humans appear in the literature as "users," "operators," "stakeholders," or "interactors"—but the formal design of roles, the specification of skills, and the creation of certification pathways that are structurally required in AI-intensive environments are rarely foregrounded.

This omission has practical consequences. When organisations deploy AI systems without explicit attention to role design, the default outcome is often the erosion or fragmentation of existing jobs. Tasks that can be automated are stripped away, leaving workers with residual responsibilities that may be less coherent, less meaningful, and less valued. The potential for creating new, elevated roles rooted in human strengths—judgment, contextual reasoning, ethical discernment, care—remains unrealised because no systematic framework exists for designing such roles. This paper addresses that gap.

1.5 Research Questions

We investigate the following research questions:

- RQ1: How can HITL and XAI capabilities be structured so that human participation is necessary and valuable for organisational performance, safety, and fairness?
- RQ2: How can these capabilities be translated into explicit role families that reinstate and upgrade jobs in AI-intensive domains?
- RQ3: What competency, skills, and certification frameworks are required to support access to, and progression within, human-in-the-loop roles?
- RQ4: Which organisational and public-policy governance mechanisms are needed to embed human agency, rights, and recognition within AI-mediated labour systems, and to elevate human capabilities and quality of life?

1.6 Contributions

This paper offers four contributions to scholarship and practice.

First, we synthesise labor-aware requirements for AI design, drawing on research on AI and the future of work, HITL, XAI, human-centered AI, fairness, and well-being to derive requirements for deployments that explicitly account for job structures, skills, and human flourishing. Second, we

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introduce a human-in-the-loop labor architecture framework—a four-layer structure of capabilities, roles, skills and certification, and governance—that connects AI system affordances to role design, skill development, and institutional arrangements. Third, we propose concrete job families and role archetypes in automation, HR, collaboration, and well-being that rely structurally on HITL and XAI, showing how jobs can be reinstated and elevated rather than simply displaced. Fourth, we outline skills, certification, and policy pathways through which governments and institutions can support these roles, using AI to raise the baseline of human capabilities and quality of life.

2 Methodology and Approach

This is a conceptual, theory-building paper grounded in an integrative literature review and architectural modelling. The approach proceeds in four phases.

2.1 Narrative Synthesis

First, we conduct a narrative synthesis of research across multiple intersecting domains: AI and the future of work, human-in-the-loop methods, explainable AI, human-AI interaction, human-centered AI, fairness in organisational AI, and workplace well-being. The synthesis is integrative rather than systematic in the formal sense; we aim to identify core concepts, debates, and design dimensions that can inform the construction of a labor-architecture framework.

Tsiakas and Murray-Rust's analysis of HITL and XAI in future work practices serves as an anchor for this synthesis. They identify four salient domains (automation, HR, collaboration, human factors) and four key design aspects (system design, roles and interaction types, autonomy and control, learning and personalisation). We use this structuring as a starting point while extending it toward labor-specific considerations.

2.2 Abductive Conceptual Development

Second, we engage in abductive conceptual development, iteratively moving between the literature and emerging constructs. Abduction, in this context, involves reasoning from observed phenomena (the capabilities and interaction patterns documented in HITL/XAI research) to plausible explanatory frameworks (the labor architectures that could make those patterns sustainable and meaningful).

A critical move in this abductive process is the reframing of existing design aspects. Where much HITL/XAI research asks, "How should humans and AI interact?" we shift the question to, "Which roles, skills, and institutions are required for those interactions to support meaningful work?" This reframing opens space for thinking about jobs, careers, and labour markets as integral to AI system design rather than as external outcomes.

2.3 Architectural Framework Construction

Third, we construct an architectural framework organised into four layers. The architectural metaphor is deliberate: it suggests that sustainable human-AI work systems require coherent

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structuring across multiple levels, from technical affordances to institutional arrangements. Each layer is defined by its constituent elements and by its relationships to adjacent layers.

The capability layer captures the technical and interaction affordances required for meaningful HITL work. The role layer defines job families and responsibilities built upon those capabilities. The skills and certification layer specifies competencies and credentialing arrangements enabling access to those roles. The governance layer articulates organisational and policy mechanisms that institutionalise human agency and rights.

2.4 Domain-Specific Illustrations

Finally, we develop domain-specific illustrations in automation, HR, collaboration, and well-being. These illustrations take the form of role archetypes and skills pathways inspired by existing practice and research, but not tied to any specific empirical case. They are structured thought experiments intended to demonstrate the framework's applicability and to guide subsequent empirical studies.

The illustrations are deliberately generative rather than prescriptive. Different organisations, sectors, and national contexts will require different instantiations of the framework. Our aim is to provide a vocabulary and a set of design dimensions that can support context-sensitive adaptation.

3 Background: AI, Work, and Human-Centered Design

3.1 AI, Automation, and Employment

The relationship between AI and employment has been extensively analysed from economic, sociological, and organisational perspectives. Manyika and Sneader provide a comprehensive assessment of automation's potential impacts, identifying both the scale of potential disruption and the critical role of complementary investments in skills and institutions. They argue that employment outcomes are highly sensitive to policy choices rather than predetermined by technology—a finding that underscores the importance of deliberate design in shaping how AI affects work.

Dantas et al. describe both benefits and hurdles of AI in the workplace. On the benefit side, they identify potential productivity gains, improved decision-making through data-driven insights, and the emergence of new forms of work that combine human and machine capabilities. On the hurdle side, they document concerns about trust, ethics, and acceptance, noting that workers' willingness to engage with AI systems depends crucially on whether those systems are perceived as fair, transparent, and supportive of human goals.

In human resources, predictive models have been deployed to estimate job performance and competence using factors such as job knowledge, self-motivation, and role perception. These models promise more efficient—and potentially more objective—screening, but they also raise concerns about bias, transparency, and the reduction of complex human qualities to algorithmic scores. User studies with young job applicants suggest that they value the efficiency and perceived impartiality of AI-enabled recruiting, while remaining sceptical of systems that appear biased, unpredictable, or invisible in their workings.

In manufacturing, AI and smart technologies have been deployed to automate human-like and

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repetitive tasks, leading to concerns about job displacement and deskilling. However, recent work argues for HITL approaches that sustain human meta-learning capabilities and better integrate human and non-human actors. These approaches recognise that human flexibility, contextual knowledge, and ethical judgment remain valuable even in highly automated environments.

3.2 Human-in-the-Loop and Explainable AI

Human-in-the-loop methods integrate human feedback into AI pipelines at multiple stages. During development, humans may label training data, provide evaluative feedback on model outputs, or guide learning processes through interactive demonstrations. During deployment, humans may intervene in decisions, override automated actions, or provide ongoing feedback that shapes model adaptation.

The scope of HITL engagement varies across applications. In some cases, humans are involved intermittently, stepping in only when the system encounters uncertainty or high-stakes situations. In others, human involvement is continuous, with real-time feedback shaping system behaviour moment by moment. van Berkel, Skov, and Kjeldskov distinguish between intermittent, continuous, and proactive forms of human-AI interaction, each with different implications for role design and skill requirements.

Explainable AI techniques aim to render AI behaviour intelligible and accountable. These techniques range from local feature attributions (explaining why a particular decision was made) to global model descriptions (documenting overall behaviour and limitations). Model cards provide structured documentation of model purpose, performance characteristics, and known limitations. Suresh et al. emphasise that explanations must be tailored to stakeholder goals, cognitive capacities, and the decisions at stake—a finding with direct implications for role design.

Theory-driven, user-centric XAI frameworks argue that explanations are not just technical outputs; they are communicative acts embedded in social and organisational contexts. What counts as a good explanation depends on who is asking, what they need to understand, and what actions they might take based on that understanding. Tsiakas and Murray-Rust synthesise HITL and XAI research into a design space for future work practices, highlighting aspects such as input–output specification, user roles, autonomy and control, and learning and personalisation.

3.3 Fairness, Well-Being, and Human-Centered AI

Research on fair AI in employee management examines how algorithmic systems influence distributive, procedural, and interactional fairness. Distributive fairness concerns the outcomes of decisions—whether resources, opportunities, and burdens are allocated equitably. Procedural fairness concerns the processes by which decisions are made—whether they are transparent, consistent, and open to input. Interactional fairness concerns the quality of interpersonal treatment—whether people are treated with dignity and respect.

Design agendas for fair organisational AI emphasise transparency, user voice, and contestability. Transparency ensures that affected individuals can understand how decisions are made. User voice provides mechanisms for input and feedback. Contestability enables individuals to challenge decisions they believe are erroneous or unfair. Shneiderman's human-centered AI framework calls

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for combinations of high automation and high human control, with the goal of reliable, safe, and trustworthy systems that people can understand and appropriately influence.

Studies of AI-mediated collaboration show that real-time feedback on participation and influence can shape group behaviour and support soft skill development. However, such feedback must be carefully designed to avoid becoming a source of stigma or bias. When AI systems monitor and visualise team dynamics, they create new forms of visibility that can be used for development or for surveillance—a distinction that depends on governance and role design. Human factors research documents the mental workload and stress generated by digital technologies, emphasising the need to balance efficiency with well-being. When AI systems add to cognitive load without providing compensating benefits, they can degrade rather than enhance work experience.

3.4 The Gap: From Interaction to Labor Architecture

Across these literatures, a consistent pattern emerges: the technical design of AI systems, the interaction patterns they support, and the governance arrangements surrounding them all shape how work is experienced and valued. Yet the link to formal job structures, skill requirements, and certification pathways remains underdeveloped.

In practice, organisations deploying AI systems face decisions about job design: Which tasks will be automated? Which will remain with humans? What new tasks will emerge? What skills will be needed? How will those skills be developed and recognised? Without frameworks to guide these decisions, organisations default to patterns that may not serve human flourishing or long-term organisational effectiveness. The concept of human-in-the-loop labor architectures is intended to fill this gap by providing a systematic framework for linking AI capabilities to role design, skill development, and institutional governance.

4 Conceptual Framework: Human-in-the-Loop Labor Architectures

4.1 Definition

We define a human-in-the-loop labor architecture as a structured set of AI system capabilities, human roles, skill and certification pathways, and governance mechanisms that together ensure that:

1. Human judgment, contextualisation, and ethical reasoning are structurally necessary for AI-mediated processes. The system is designed such that its safe, fair, and effective operation depends on ongoing human participation.
2. Workers have clearly defined, recognised, and supported roles in shaping AI behaviour and outcomes. These roles are formalised in job descriptions, performance expectations, and career pathways.
3. Skills and credentials enable broad access to these roles and mobility across sectors. Education and training systems prepare workers for HITL roles, and certification schemes validate their competencies.

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4. Organisational and public-policy instruments secure human oversight, rights, and recognition. Governance mechanisms at multiple levels ensure that human-centered arrangements are not merely aspirational but enforced and sustained.

This concept reframes AI from a purely automating force to a technology whose deployment can be designed to support role complementarities and job reinstatement at higher levels of responsibility and skill. The question is not simply "What can AI do?" but "How can AI and humans be configured together to create value, support dignity, and enable flourishing?"

4.2 Theoretical Foundations

The framework draws on several theoretical traditions.

Socio-technical systems theory emphasises that work systems include both technical and social elements, and that they need to be designed together. Optimal performance depends on alignment between technical capabilities and social arrangements, including roles, skills, and norms. HITL labor architectures extend this insight to AI-intensive environments by treating human roles as integral to system functioning rather than residual. Labour process theory examines how work is organised and controlled, highlighting how technology can deskill or upgrade work and how worker agency shapes these outcomes; our framework is oriented toward upgrading through deliberate design. Institutional theory focuses on how formal and informal rules shape organisational behaviour, which is reflected in our governance layer. The capabilities approach, associated with Sen and Nussbaum, focuses on what people are able to do and to be—their real freedoms to achieve functionings they have reason to value—and informs our emphasis on skills, certification, and governance as ways to protect and expand human flourishing.

4.3 The Four Layers

The framework is structured into four layers, each building on the one below:

- Capability layer: Technical and interaction affordances required for meaningful HITL work.
- Role layer: Job families, responsibilities, and career pathways built on HITL capabilities.
- Skills and certification layer: Competency domains, training pathways, and credentialing systems that enable access to HITL roles.
- Governance layer: Organisational policies, regulation, and public policy that enforce and support human-centered arrangements.

The layers are recursively related: capabilities enable roles, roles require skills, skills are certified through institutional mechanisms, and governance shapes all layers. Changes in any layer may propagate through the system.

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5 Capability Layer: Affordances for Human Co-Performance

The capability layer comprises the technical and interaction affordances that AI systems must provide to enable meaningful HITL work. These are not merely features that are nice to have; they are structural requirements for architectures in which human participation is necessary and valuable.

5.1 Multi-Perspective Intelligibility

For humans to exercise meaningful oversight, they must be able to understand what AI systems are doing and why. This requires intelligibility at multiple levels and from multiple perspectives.

Global transparency involves documentation of model purpose, data sources, development processes, performance metrics, and known limitations. Local explanations provide case-specific rationales for individual outputs, tailored to user roles and goals. Communication of uncertainty and limitations enables humans to calibrate their trust and to know when intervention is most needed. Without such intelligibility, human actors cannot responsibly accept, challenge, or improve AI-supported decisions; they are reduced to either blind compliance or arbitrary rejection.

5.2 Granular Intervention and Control

Intelligibility alone is insufficient; humans must also be able to act on their understanding. Interfaces must allow humans to exercise meaningful control over AI operations at appropriate levels of granularity.

Decision-level interventions include confirming or rejecting AI recommendations. Parameter-level interventions involve adjusting thresholds or parameters within governance-approved ranges. Process-level interventions include pausing automation and switching to manual workflows when systems encounter situations they cannot handle reliably or when stakeholders raise concerns. Escalation mechanisms enable multi-party reviews for contentious or high-risk cases. These interventions must be technically binding and appropriately logged to support accountability and learning.

5.3 Structured Feedback and Annotation

HITL roles involve not only intervening in AI decisions but also providing feedback that shapes future system behaviour. Systems must capture human feedback in ways that can be used for both immediate decision-making and long-term learning.

Evaluative feedback includes ratings and judgments about AI outputs. Corrective feedback includes alternative decisions or revised labels that directly improve outputs. Contextual annotations provide information about local conditions and norms that may not be captured in training data. Disagreement signals from different actors highlight areas where AI and human judgments diverge, indicating potential problems with model design or deployment context.

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5.4 Contribution Traceability and Impact Analytics

For HITL roles to be valued and sustained, the contributions of human workers must be visible and demonstrably valuable. Systems must track and analyse human contributions in ways that support recognition, learning, and role development.

Activity metrics include counts and types of interventions over time. Impact metrics track changes in performance, fairness, or safety associated with human interventions. Pattern analytics identify where and when human judgment diverges from AI outputs and under what conditions it adds the most value. Contribution traceability must be designed to support learning and recognition, not surveillance or control.

6 Role Layer: Reinstated and Reconfigured Jobs

The role layer translates capabilities into job families and role definitions. We focus on "reinstated" and "reconfigured" roles—jobs that gain new relevance when AI is deployed with a HITL labor architecture, and that might otherwise be displaced or downgraded. Following Tsiakas and Murray-Rust, we organise examples around four domains: automation, human resources, collaboration, and well-being. For each domain, we propose role archetypes that illustrate how HITL capabilities can be embedded in formal job structures.

6.1 Automation and Smart Manufacturing

6.1.1 AI Line Steward

The AI Line Steward is responsible for overseeing AI-controlled production lines, recognising that even sophisticated automation systems require human judgment to handle edge cases, local conditions, and unexpected events.

Responsibilities include monitoring AI control systems via explanation dashboards and alerts, interpreting signs of uncertainty or anomalies, deciding when to accept AI actions versus override or pause operations, annotating events with contextual information, and participating in post-incident reviews and continuous improvement processes. Experienced production operators at risk of displacement can be retrained and certified into this role, shifting from manual execution to oversight and interpretation. Required capabilities include multi-perspective intelligibility, granular intervention and control, structured feedback, and contribution traceability.

6.1.2 Human-Centered Safety Engineer

The Human-Centered Safety Engineer focuses on safety-critical AI behaviour in automated environments.

This role reviews logs of safety-related AI decisions, examines patterns in near misses and incidents, coordinates with control engineers to adjust safety thresholds, incorporates worker feedback about unsafe or unexpected AI behaviour, leads investigations of safety incidents, and participates in

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design reviews for new or updated AI systems. Traditional safety and quality roles are reconfigured to explicitly include AI behaviour, with domain expertise becoming essential for ensuring safe real-world operation. The role relies on intelligibility of safety-relevant AI behaviour, granular intervention, structured feedback, and traceability of safety analyses and interventions.

6.1.3 Human-Machine Collaboration Specialist

The Human-Machine Collaboration Specialist designs and supports collaborative arrangements between humans, robots, and AI systems, including cobots that work alongside human operators.

Responsibilities include analysing workflows to allocate tasks between humans and machines, configuring collaborative robots and AI assistants, training workers in effective collaboration, collecting feedback on collaboration experiences, and identifying opportunities for deeper integration that enhances productivity and worker experience. This reconfigured role combines industrial engineering, human factors, and social science and becomes more important as robots and AI systems grow more capable.

6.2 Human Resources and Talent Governance

6.2.1 Algorithmic Hiring Steward

The Algorithmic Hiring Steward manages AI-enabled recruitment and promotion systems with a focus on fairness, transparency, and accountability.

The steward monitors predictive models used in screening and ranking, interprets explanations of ranking decisions, responds to candidate inquiries and contestations, feeds fairness concerns and contextual information back into system design, and participates in audits and bias-mitigation efforts. HR generalists and recruiters can be upgraded into this role, moving from manual screening to stewardship of AI systems and stakeholder relations. The role depends on intelligibility, granular intervention, structured feedback, and contribution traceability.

6.2.2 AI-Aware Career Path Designer

The AI-Aware Career Path Designer uses explainable competence and performance models to support employee development and internal mobility.

Responsibilities include translating model-based insights into transparent capability frameworks, helping workers understand how skill acquisition and performance affect AI-predicted outcomes and trajectories, co-designing training and mobility paths, monitoring unintended consequences such as self-fulfilling prophecies, and advocating for employees when AI assessments miss context. Career development professionals are reconfigured from administering standardised processes to interpreting AI-generated insights in human-centered ways.

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6.2.3 People Analytics Ethicist

The People Analytics Ethicist provides ethical oversight of employee data and AI-based analytics.

This role defines principles and guidelines for data collection and use, reviews analytics projects for compliance with those principles, assesses risks related to privacy and autonomy, engages with employee representatives and other stakeholders, and recommends design and governance changes to mitigate harms. Traditional HR compliance or legal roles can be upgraded into this position, with a stronger focus on socio-technical judgment and cross-disciplinary knowledge.

6.3 Collaboration and Team Dynamics

6.3.1 Team Insights Facilitator

The Team Insights Facilitator manages AI-mediated feedback about participation, communication, and influence in teams.

Responsibilities include configuring AI tools that visualise participation and interactions, interpreting feedback for teams, coaching members on how to use insights constructively, monitoring for problematic effects such as stigma or gaming, and advocating changes to tool design or governance where needed. Roles in organisational development and facilitation can be upgraded into this archetype, anchoring human mediation and judgment in AI-mediated collaboration.

6.3.2 Collaboration Pattern Curator

The Collaboration Pattern Curator analyses AI-generated patterns of coordination, communication, and knowledge sharing across teams.

The curator identifies best-practice collaboration patterns, documents them as reusable templates, supports teams in adopting and adapting patterns, and works with tool designers to create pattern-aware features. Knowledge management or process improvement roles can be reconfigured to focus on socio-technical collaboration patterns rather than static procedures.

6.3.3 Boundary-Spanning Sensemaker

The Boundary-Spanning Sensemaker uses AI-supported analytics to interpret cross-team or cross-organisational collaboration, especially in complex ecosystems.

Responsibilities include integrating AI-generated insights with qualitative understanding, producing narratives that make sense of interdependencies and risks, and supporting strategic decision-making about alliances, platforms, and innovation networks. This role builds on and upgrades strategic analyst or ecosystem manager positions, focusing on sensemaking rather than aggregate metrics.

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6.4 Well-Being and Human Factors

6.4.1 Digital Well-Being Steward

The Digital Well-Being Steward oversees AI systems that monitor and influence workload, stress, and engagement.

Responsibilities include configuring well-being analytics, interpreting signals in dialogue with workers, ensuring that interventions are supportive rather than punitive, and escalating structural issues such as staffing or workload design. Roles in occupational health, HR, or employee support services can be upgraded into this archetype, combining human care with AI-based sensing.

6.4.2 Context Narrator

The Context Narrator ensures that AI systems processing behavioural or performance data are informed by contextual information that would otherwise be invisible.

This role collects and documents contextual factors influencing data patterns, curates narratives that accompany quantitative indicators, and collaborates with data scientists and managers to avoid misinterpretation. It reconfigures roles in line management, union representation, or employee advocacy toward structured contextualisation of AI-mediated assessments.

6.4.3 Workload Balancing Specialist

The Workload Balancing Specialist uses AI tools to anticipate overload, bottlenecks, and burnout risks and to design more sustainable patterns of work.

Responsibilities include working with AI-driven workload forecasts, discussing options with affected workers and managers, adjusting schedules and task allocation, and feeding insights back into system design and organisational policy. Existing roles in operations planning, HR planning, or team leadership can be upgraded into this function, anchoring human negotiation and judgment in AI-based workload analytics.

7 Skills and Certification Layer: Pathways into Elevated Roles

The skills and certification layer identifies the competency domains and credentialing arrangements that support access to HITL roles and mobility across sectors. It links individual development to the structural requirements of human-in-the-loop labor architectures.

7.1 Core Competency Domains

Four cross-cutting competency domains are central:

- AI literacy and interpretive skill: Understanding AI capabilities and limits, reading explanation outputs, and recognising uncertainty and potential bias.

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- Domain expertise and contextual judgment: Deep knowledge of the relevant domain (e.g., manufacturing, HR, collaboration, well-being), including its norms, risks, and values.
- Socio-ethical reasoning: Recognising fairness, privacy, and autonomy issues and reasoning through trade-offs and stakeholder impacts.
- Interaction and facilitation skills: Communicating about AI-mediated decisions, facilitating dialogue, and supporting constructive engagement with feedback and analytics.

These domains are instantiated differently across role families but provide a common foundation for HITL labor architectures.

7.2 Role-Specific Skill Profiles

Each role archetype corresponds to a role-specific skill profile that combines the four core domains with technical or procedural skills. For example, an AI Line Steward requires familiarity with control systems and production processes, while a People Analytics Ethicist needs stronger grounding in data protection law and organisational ethics.

Skill profiles can be expressed as competency matrices that specify levels (basic, intermediate, advanced) for each domain and associated micro-skills, enabling modular training and stackable credentials. This modularity supports entry from different backgrounds and progression across roles and domains.

7.3 Certification Pathways and Stackable Credentials

Certification pathways make HITL roles visible and portable in labour markets. They can include:

- Short-course certificates focused on AI literacy and interpretability.
- Domain-specific HITL role certificates (e.g., Certified AI Line Steward, Certified Algorithmic Hiring Steward).
- Advanced diplomas or professional master's programmes in human-centered AI and labor architectures.

Stackable credentials allow workers to build from foundational literacy to specialised roles over time, with recognition of prior learning and experience. Public institutions and industry bodies can co-develop standards so certifications are trusted across organisations and sectors.

7.4 Inclusion, Access, and Career Mobility

Designing skills and certification pathways for inclusion is critical. Entry-level HITL roles can be made accessible to workers displaced or at risk from automation, with subsidised training and recognition of tacit knowledge.

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Career ladders can be designed so that workers can progress from operational HITL roles into design, governance, or policy roles, expanding the diversity of perspectives in decision-making about AI systems. This supports both individual mobility and the diffusion of human-centered practices across the economy.

8 Governance Layer: Organisational and Policy Backing

The governance layer comprises organisational policies, regulatory frameworks, and public-policy instruments that institutionalise HITL labor architectures and protect human agency and rights.

8.1 Organisational Governance

Within organisations, governance mechanisms include:

- Policies that define which decisions must involve human oversight and at what level.
- Procedures for documenting and auditing AI-mediated decisions and their human interventions.
- Mechanisms for worker participation in the design and monitoring of AI systems, such as joint committees or consultative bodies.

These arrangements ensure that HITL roles are not purely nominal but are embedded in decision rights, accountability structures, and performance metrics.

8.2 Regulatory Frameworks

Regulators can support HITL labor architectures by specifying requirements for human oversight in high-risk AI applications, mandating transparency and contestability, and recognising certified HITL roles in compliance regimes.

For example, regulation might require that certain AI-mediated decisions (e.g., hiring, credit, safety-critical control) be reviewable by qualified humans with documented authority to override or modify decisions and that organisations maintain records of such interventions.

8.3 Public Policy Instruments

Public policy can use funding, incentives, and public procurement to promote human-centered architectures. Instruments include:

- Subsidies or tax credits for training and certification in HITL roles.
- Grants for experimentation with human-centered AI and labor architectures in public services.
- Procurement criteria that favour systems designed with HITL capabilities and recognised human roles.

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By shaping demand and supporting capacity building, policy can steer AI adoption toward human elevation rather than displacement.

8.4 Social Dialogue and Collective Representation

Social partners—unions, employer associations, professional bodies, and civil society—play a crucial role in shaping governance.

Collective agreements can include provisions on AI use, HITL roles, training rights, and protections against surveillance and unfair automated assessments. Professional bodies can define ethical standards and certification requirements for roles such as People Analytics Ethicists or Algorithmic Hiring Stewards.

9 Discussion and Research Agenda

The proposed framework opens a research agenda on how HITL labor architectures can be designed, implemented, and scaled in different contexts.

9.1 Empirical Studies of Emerging HITL Roles

Empirical research is needed to document and analyse emerging HITL roles in practice, including how they are created, how they function, and what challenges they face. Case studies across sectors can reveal patterns of successful and unsuccessful role design.

Such studies can also explore workers' experiences of these roles, including issues of recognition, workload, autonomy, and career prospects, helping to refine role archetypes and skill profiles.

9.2 Design Experiments and Prototyping

Design experiments and prototyping can test different configurations of capabilities, roles, and governance in controlled settings.

Living-lab approaches in organisations or public services can experiment with new HITL roles, track impacts on performance and well-being, and iterate on design. Collaboration between researchers, practitioners, and workers is crucial for co-creating viable architectures.

9.3 Measurement and Evaluation Frameworks

New measurement frameworks are needed to evaluate HITL labor architectures.

Beyond traditional performance metrics, evaluations should include indicators of human flourishing, fairness, inclusion, and capability expansion. Contribution traceability and impact analytics at the capability layer can feed into these broader assessments, linking micro-level interventions to macro-level outcomes.

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9.4 Comparative and Institutional Analysis

Comparative studies can examine how different institutional contexts—labour market regimes, welfare states, regulatory traditions—shape the emergence and sustainability of HITL labor architectures.

Institutional analysis can identify enabling and constraining factors, informing policy strategies that are sensitive to local conditions while aligned with shared human-centered principles.

10 Conclusion

10.1 Synthesis of Contributions

This paper has argued that the future of work with AI will be shaped not only by what systems can technically do, but by how we design the labor architectures around them. By articulating a framework of human-in-the-loop labor architectures—spanning capabilities, roles, skills and certification, and governance—we have shown how HITL and XAI can be used to reinstate and reconfigure jobs, support skill elevation, and create institutional pathways for human flourishing.

Our four contributions to scholarship and practice are as follows. First, we synthesised labor-aware requirements for AI design, drawing on research across multiple fields to identify what AI systems must make possible for meaningful human participation: multi-perspective intelligibility, granular intervention and control, structured feedback and annotation, and contribution traceability with impact analytics. Second, we introduced a four-layer framework that links AI system affordances to role design, skill development, and institutional arrangements, providing a vocabulary and structure for thinking systematically about how AI and human work can be configured together.

Third, we proposed concrete role archetypes across four domains—automation, human resources, collaboration, and well-being—that rely structurally on HITL and XAI, illustrating how jobs can be reinstated and elevated rather than simply displaced, turning potential job losses into opportunities for upgraded roles. Fourth, we outlined competency domains, certification schemes, and public-policy instruments through which governments and institutions can support these roles, showing how the framework can be operationalised in practice by specifying pathways for skill development and mechanisms for governance.

10.2 Implications for Practice

For different stakeholders, the framework suggests specific implications. For organisations, the recommendation is to treat AI deployment as an opportunity to design new, elevated roles rather than merely to reduce headcount, investing in the capabilities, training, and governance arrangements that make HITL labor architectures possible, and recognising that sustainable AI deployment requires ongoing human participation supported through appropriate role design, skill development, and compensation.

For technology developers, the implication is to design AI systems with HITL capabilities from the outset, not as afterthoughts, by building intelligibility, intervention, feedback, and traceability into system architectures and working with domain experts and HITL practitioners to understand

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what capabilities are needed in practice. For workers and their representatives, the framework invites proactive engagement with AI deployment, advocacy for HITL labor architectures that create elevated roles and protect worker interests, participation in governance bodies and certification development, and the pursuit of skills for HITL roles through available pathways.

For educators and trainers, the implication is to develop programmes that prepare workers for HITL roles by integrating AI literacy, domain expertise, ethics, and facilitation, partnering with employers and certification bodies to ensure programmes meet real needs, and creating accessible pathways for diverse learners. For certification bodies, the framework suggests developing robust, credible certification schemes co-created with stakeholders, ensuring accessibility and portability, and maintaining quality assurance while avoiding unnecessary barriers. For policymakers, the implications include establishing minimum standards for human oversight, explanation, and contestation rights; providing funding and incentives for HITL training and certification; using public procurement to encourage adoption; supporting research and innovation; and engaging internationally to develop standards and cooperation.

10.3 A Vision for the Future

The framework presented in this paper is motivated by a particular vision of the future: one in which AI is used not to minimise human involvement but to create spaces where human judgment, creativity, and care are central to value creation and social well-being. In this vision, workers are elevated to roles of stewards, interpreters, and co-creators in socio-technical systems; they are not residual elements to be eliminated when technically feasible, but essential participants whose contributions are recognised, valued, and supported.

This vision is not inevitable. It requires deliberate design, sustained investment, and ongoing governance, as well as choices—by organisations, by workers, by policymakers—about what kinds of work we value and what futures we want to create. The alternative is a future in which AI deployment proceeds without attention to human roles, eroding jobs, concentrating power, and diminishing human agency. The choice is not between AI and no AI; the choice is between different ways of configuring AI with human work. Human-in-the-loop labor architectures offer one way—a way that reinstates and elevates work, invests in skills and certification, and governs technology in the service of human flourishing; realising this vision will require effort, imagination, and collective action, but the possibility is there, waiting to be built.

10.4 Closing Reflection

In 2018, Manyika and Sneader observed that employment outcomes from AI and automation would be highly sensitive to policy choices rather than predetermined by technology, and that observation remains true today, with implications that have only grown more urgent. As AI systems become more capable and pervasive, the choices we make about their design and deployment will shape work for generations.

The framework offered here is one contribution to a larger conversation about those choices. It is not a blueprint to be applied mechanically, but a set of concepts and structures to be adapted, tested, and refined; it invites empirical investigation, critical engagement, and practical experimentation, and most importantly, it invites us to think about AI and work not as something that happens

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to us, but as something we can shape. The future of work with AI will be built; the question is whether we build it deliberately, with attention to human flourishing, or whether we let it build itself, with whatever outcomes emerge. Human-in-the-loop labor architectures are a proposal for deliberate construction—a way of building that keeps people at the centre.

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